F1 Race Performance Analysis

Goal: compare driver and constructor performance and explore how starting grid position relates to finishing position and race points.

Dataset: 4,219 driver-race entries across 10 seasons (2015–2024), 209 races, 53 drivers and 18 constructors. One row represents one driver's result in one race.

Inputs: races.csv, drivers.csv and race_results.csv in 00-Raw_data. Results are joined to drivers on driverId and to races on season and round. The original external dataset source is not documented in the repository; the team must add its attribution before submission.

Numerical variables: points, grid, position and laps. Categorical variables: driver, constructor, nationality and result status. Analysis uses the team's cleaned CSV.

Preparation: the existing cleaning script keeps seasons from 2015 onward, removes duplicate rows, fills missing numeric values with medians and missing text with Unknown, and converts selected dates and numbers. Some date conversions may still produce missing values.

Scope: totals describe this dataset, not a complete official championship ranking. Cumulative points depend on participation. Grid analyses use the same selected statuses as the team EDA, exclude grid 0, and show associations rather than causal effects.

Numerical Summary Statistics

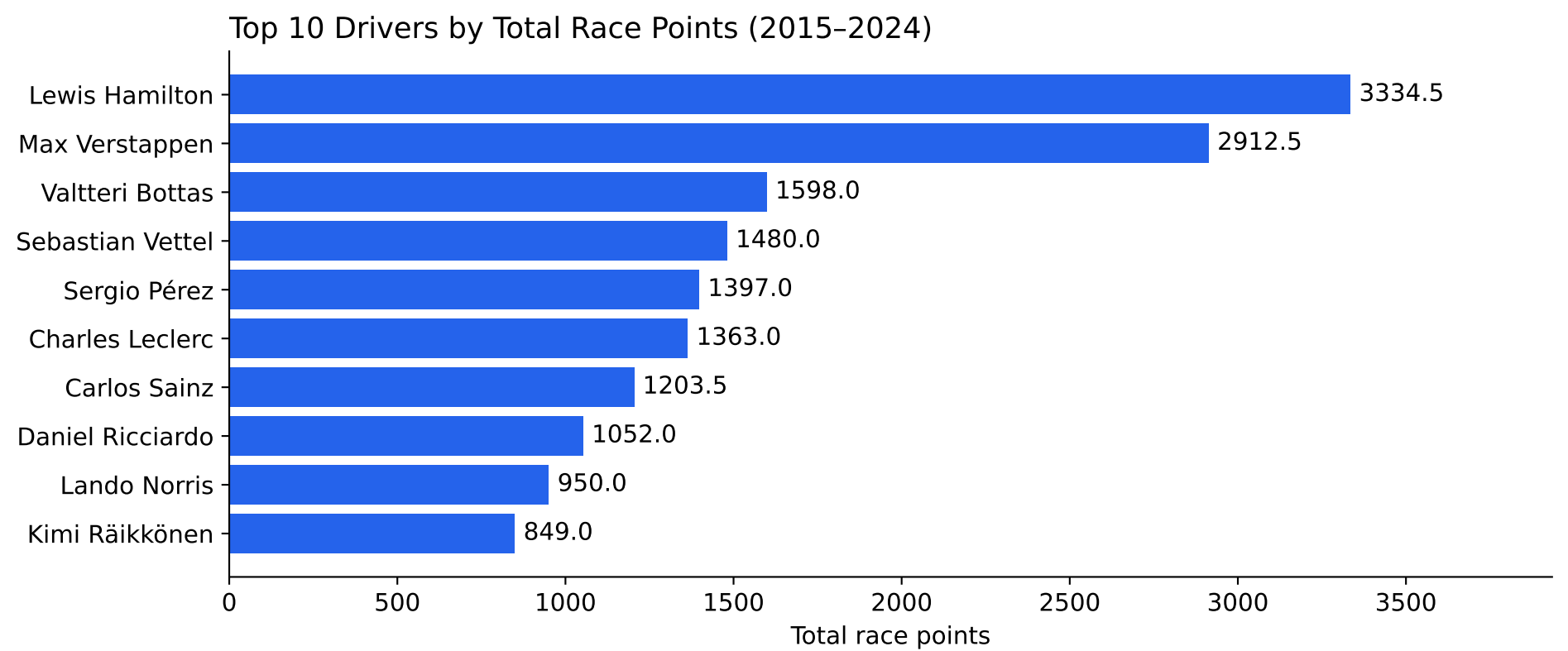
Statistic	points	grid	position	laps
count	4219.0	4219.0	4219.0	4219.0
mean	5.02	10.26	10.6	53.83
std	7.19	5.89	5.84	17.95
min	0.0	0.0	1.0	0.0
25%	0.0	5.0	6.0	51.0
50%	0.0	10.0	11.0	56.0
75%	8.0	15.0	16.0	67.0
max	26.0	22.0	22.0	87.0

All cleaned entries. Grid 0 denotes a non-standard start and is excluded from the grid charts. Recorded position includes non-finishers.

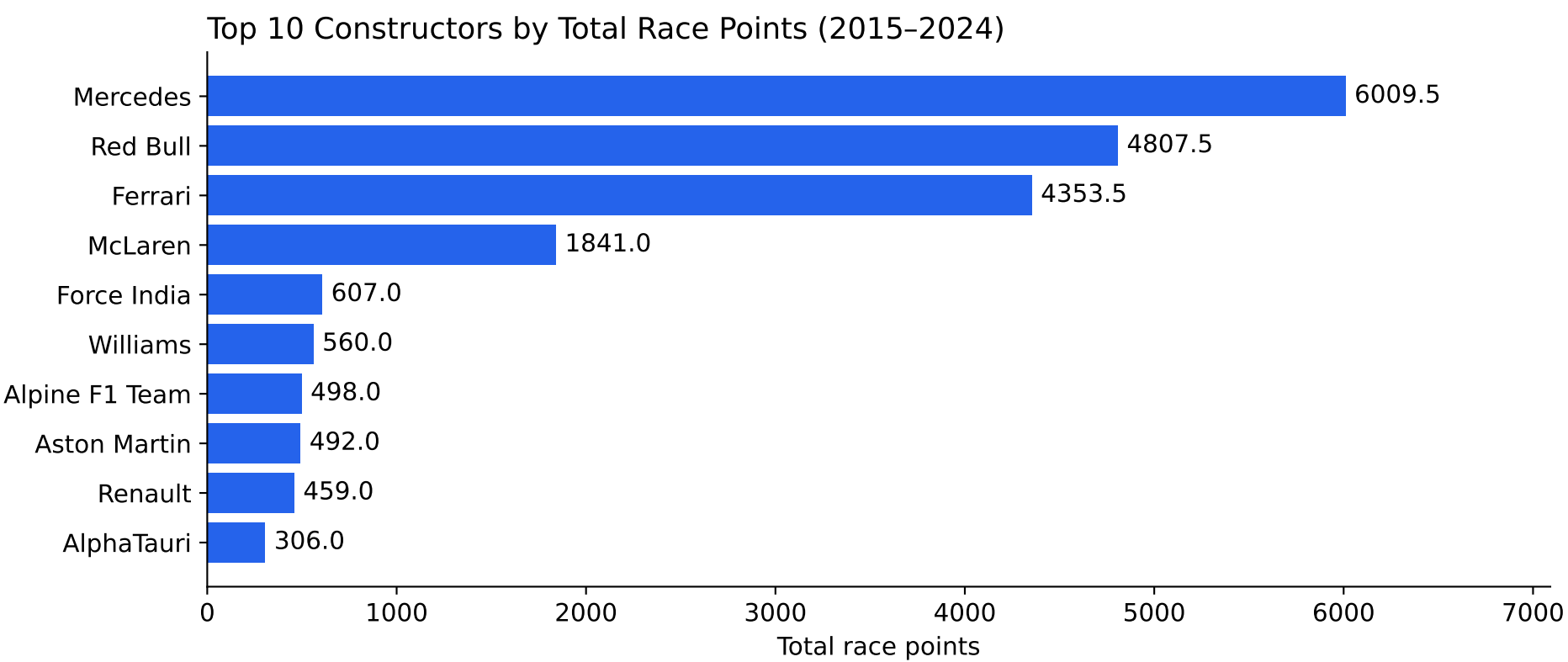
Most Frequent Result Statuses

Status	Entries	Percent
Finished	2203	52.22
+1 Lap	1083	25.67
+2 Laps	186	4.41
Collision	136	3.22
Accident	76	1.8
Engine	73	1.73
Collision damage	58	1.37
Brakes	43	1.02
Retired	42	1.0
Gearbox	38	0.9

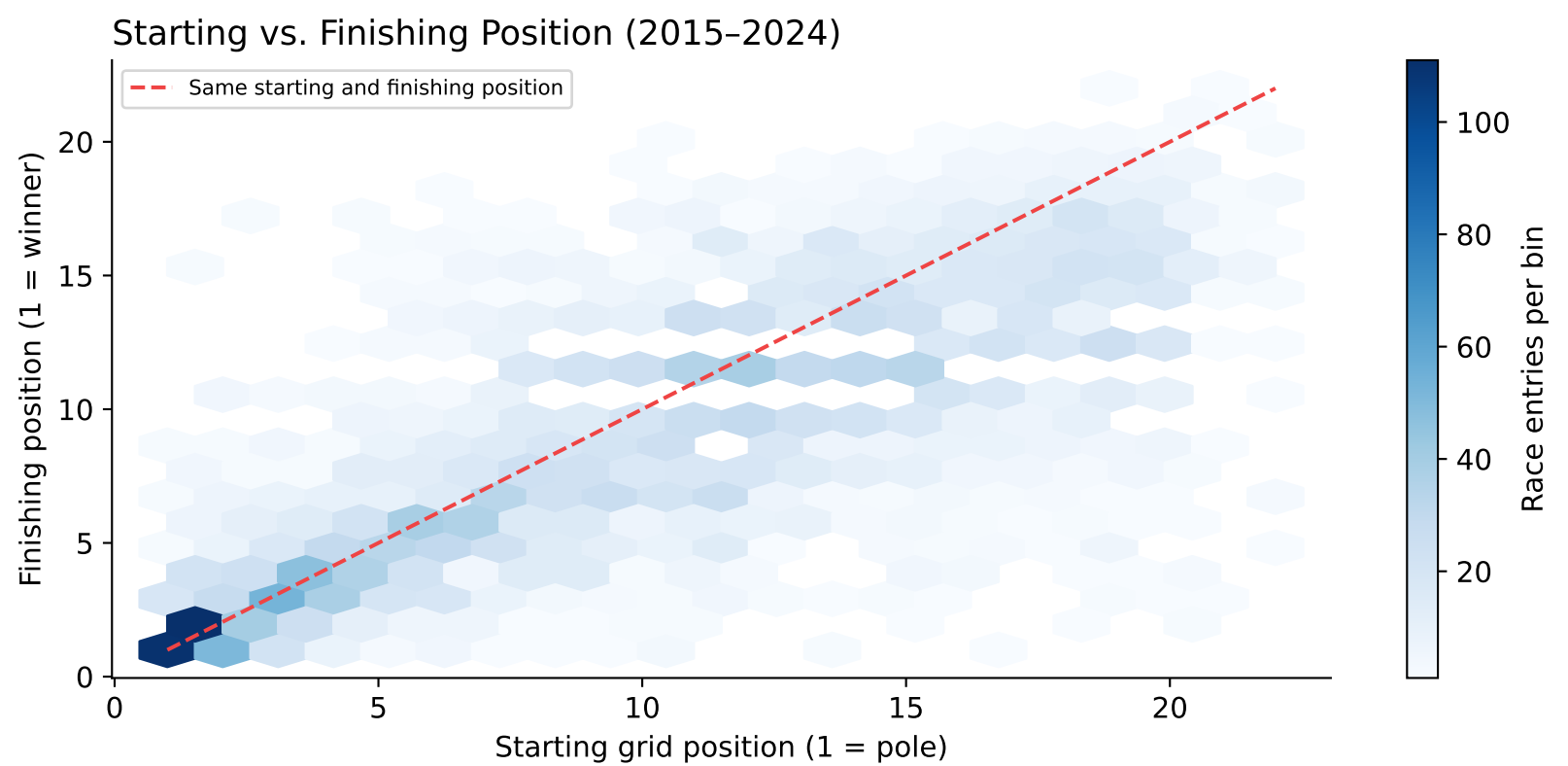
Top 10 statuses only; percentages use all cleaned driver-race entries as the denominator.



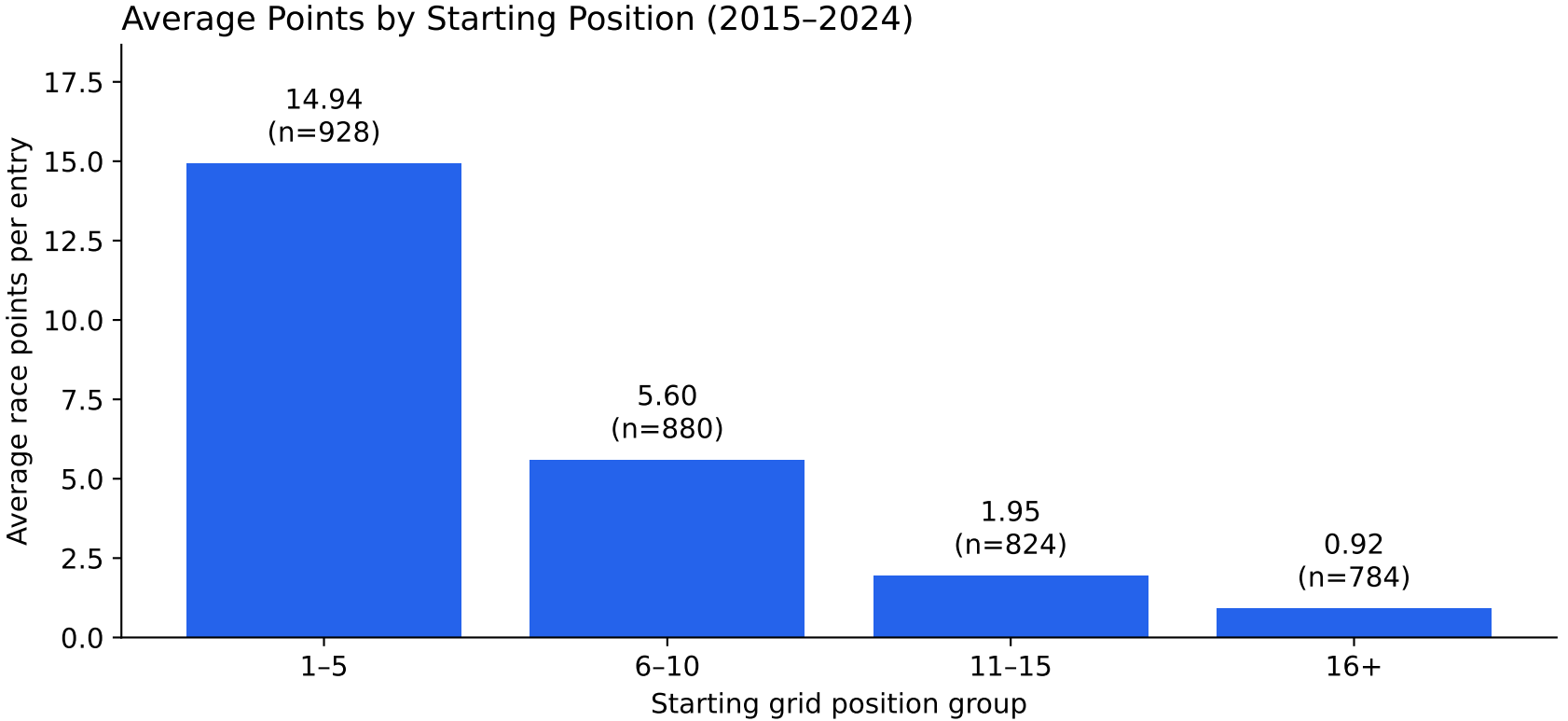
This chart ranks the ten drivers with the most accumulated race points in the 2015–2024 dataset. Longer bars indicate higher totals. These totals reflect both performance and the number of races entered, so they do not measure driver skill alone.



This chart ranks the ten constructors with the most accumulated race points in the 2015–2024 dataset. Each total combines the team's driver entries across seasons. Longer bars indicate more points, with participation and performance both contributing to the totals.



This density plot compares starting grid position on the horizontal axis with finishing position on the vertical axis. Darker hexagons represent more race entries. The dashed diagonal marks unchanged positions; entries below it gained positions, while those above it lost positions. The dark bottom-left region shows that front starters frequently finished near the front. Only entries with grid greater than zero and statuses Finished, +1 Lap, or +2 Laps are included.



This chart compares average race points for four starting-grid groups among the same selected finishers (grid > 0; statuses Finished, +1 Lap, or +2 Laps). Drivers starting 1–5 averaged 14.94 points per entry, compared with 0.92 for those starting 16 or farther back. The n labels show the number of entries in each group. Better starting positions are associated with higher average points in this subset; the chart does not establish causation.

Key Findings and Interpretation

Driver comparison: Lewis Hamilton has the highest total in this dataset with 3,334.5 race points. The chart ranks cumulative points, so longer participation can increase a driver's total.

Constructor comparison: Mercedes leads with 6,009.5 race points. Totals combine entries across seasons and should not be interpreted as performance in a single season.

Starting versus finishing: Pearson correlation is 0.761 across 3,416 selected entries. Both axes use lower numbers for better positions. Darker hexagons indicate more entries; entries below the diagonal finished in a better position than they started.

Average points by starting group: grid 1–5: 14.94 points (928 entries); grid 6–10: 5.60 points (880 entries); grid 11–15: 1.95 points (824 entries); grid 16+: 0.92 points (784 entries). These averages apply to the selected result statuses.

Categorical finding: Finished is the most frequent status, accounting for 52.22% of all cleaned entries.

Limitations: the grid charts include only Finished, +1 Lap and +2 Laps with grid > 0. Finishers further behind and non-finishers are excluded. Filtering can affect the relationship observed. Median-filled lap measurements are not used in these charts. Dataset completeness and original source attribution should be confirmed by the team.